# Example Workflow

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## Contents



## Preamble

Peak2Flux2 takes gas chromatography (GC) instrument output to calculate fluxes, specifically in greenhouse gas emissions experiments. An in development package is ready for use on GitHub. We invite you to try out the package and provide feedback to the authors of the package.

We have provided here an example workflow. Here is a data set that has measured methane (CH4) and nitrous oxide (N2O)

For questions, please reach out to:

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## Instll Peak2Flux2 and load package

```r
# Install devtools if not already installed
if (!require("devtools")) install.packages("devtools")
```

```
## Loading required package: devtools

## Loading required package: usethis
```

```r
# Load devtools
library(devtools)

# Install the GitHub package
install_github("XiaoZhangZhangRice/Peak2Flux2")
```

```
## Using GitHub PAT from the git credential store.

## Skipping install of 'Peak2Flux2' from a github remote, the SHA1 (021351c6) has not changed since las
##   Use `force = TRUE` to force installation
```

```r
# Note that this package has dependencies (i.e. requires other packages to work). When prompted, please

# Load the package
library(Peak2Flux2)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

## Fill up the excel spreadsheets

The package is designed to take a dataframe with a specific input. Please key in your raw data as per specified in the following directory.

There are two files that needs to be populated * Standard calibration file (Std_Input.csv): These are standards and their associated peak areas * Sample file (Input_samples_No_ppm.csv): These are samples and their associated peak areas, metadata used for chamber volume computation and other necessary information.

Note: There is a column called GC_Run. We define this a single continuously run, whereby a standard curve is generated and used to determine sample gas concentrations. The samples will be calibrated by the standards from the same "GC_Run". This is important and chromatography characteristics change due to considerations such as but not limited to, filter degradation, detector cleaniness (e.g. FID), reagent changes (e.g. nickel 63 in the the ECD). Hence standard curves generated for a single run are unique to that run and cannot be applied to other runs.

## Load excel files

The Standard calibration file (Std_Input.csv) and the sample file (Input_samples_No_ppm.csv) on GitHub are located in the directory: XiaoZhangZhangRice/Peak2Flux2/data

```r
#Read Standard Calibration File
std <- read.csv("../data/Std_Input.csv")

#Read Sample File
samples <- read.csv("../data/Input_samples_No_ppm.csv")
```

## Peak to PPM

This function uses data entries in the standard calibration file to create calibration curves for each individual GC run using peak areas and gas standard ppms. Afterwards, the concentration of samples (ppm) are generated from the peak areas in the samples file and the calibration curves constructed from the standard file.

```r
sample_ppm <- peak2ppm(std, samples)

sample_ppm
```

```
##                         GC_Run      Date     Plot Treatment Time_mins
## Batch12_2024.1   Batch12_2024 27-Jun-24  S-M-W-1       SMW         0
## Batch12_2024.2   Batch12_2024 27-Jun-24  S-M-W-1       SMW        21
## Batch12_2024.3   Batch12_2024 27-Jun-24  S-M-W-1       SMW        42
## Batch12_2024.4   Batch12_2024 27-Jun-24  S-M-W-1       SMW        63
## Batch12_2024.5   Batch12_2024 27-Jun-24    C-S-1        CS         0
## Batch12_2024.6   Batch12_2024 27-Jun-24    C-S-1        CS        21
## Batch12_2024.7   Batch12_2024 27-Jun-24    C-S-1        CS        42
## Batch12_2024.8   Batch12_2024 27-Jun-24    C-S-1        CS        63
## Batch12_2024.9   Batch12_2024 27-Jun-24    S-M-1        SM         0
## Batch12_2024.10  Batch12_2024 27-Jun-24    S-M-1        SM        21
## Batch12_2024.11  Batch12_2024 27-Jun-24    S-M-1        SM        42
## Batch12_2024.12  Batch12_2024 27-Jun-24    S-M-1        SM        63
## Batch12_2024.13  Batch12_2024 27-Jun-24 C-S-MD-1      CSMD         0
## Batch12_2024.14  Batch12_2024 27-Jun-24 C-S-MD-1      CSMD        21
## Batch12_2024.15  Batch12_2024 27-Jun-24 C-S-MD-1      CSMD        42
## Batch12_2024.16  Batch12_2024 27-Jun-24 C-S-MD-1      CSMD        63
## Batch12_2024.17  Batch12_2024 27-Jun-24      C-1         C         0
## Batch12_2024.18  Batch12_2024 27-Jun-24      C-1         C        21
## Batch12_2024.19  Batch12_2024 27-Jun-24      C-1         C        42
## Batch12_2024.20  Batch12_2024 27-Jun-24      C-1         C        63
## Batch12_2024.21  Batch12_2024 27-Jun-24 C-S-MD-2      CSMD         0
## Batch12_2024.22  Batch12_2024 27-Jun-24 C-S-MD-2      CSMD        21
## Batch12_2024.23  Batch12_2024 27-Jun-24 C-S-MD-2      CSMD        42
## Batch12_2024.24  Batch12_2024 27-Jun-24 C-S-MD-2      CSMD        63
## Batch12_2024.25  Batch12_2024 27-Jun-24      C-2         C         0
## Batch12_2024.26  Batch12_2024 27-Jun-24      C-2         C        21
## Batch12_2024.27  Batch12_2024 27-Jun-24      C-2         C        42
## Batch12_2024.28  Batch12_2024 27-Jun-24      C-2         C        63
## Batch12_2024.29  Batch12_2024 27-Jun-24    C-S-2        CS         0
## Batch12_2024.30  Batch12_2024 27-Jun-24    C-S-2        CS        21
## Batch12_2024.31  Batch12_2024 27-Jun-24    C-S-2        CS        42
## Batch12_2024.32  Batch12_2024 27-Jun-24    C-S-2        CS        63
## Batch12_2024.33  Batch12_2024 27-Jun-24    S-M-2        SM         0
## Batch12_2024.34  Batch12_2024 27-Jun-24    S-M-2        SM        21
## Batch12_2024.35  Batch12_2024 27-Jun-24    S-M-2        SM        42
## Batch12_2024.36  Batch12_2024 27-Jun-24    S-M-2        SM        63
## Batch12_2024.37  Batch12_2024 27-Jun-24  S-M-W-2       SMW         0
## Batch12_2024.38  Batch12_2024 27-Jun-24  S-M-W-2       SMW        21
## Batch12_2024.39  Batch12_2024 27-Jun-24  S-M-W-2       SMW        42
## Batch12_2024.40  Batch12_2024 27-Jun-24  S-M-W-2       SMW        63
## Batch12_2024.41  Batch12_2024 27-Jun-24      C-3         C         0
## Batch12_2024.42  Batch12_2024 27-Jun-24      C-3         C        21
## Batch12_2024.43  Batch12_2024 27-Jun-24      C-3         C        42
## Batch12_2024.44  Batch12_2024 27-Jun-24      C-3         C        63
## Batch12_2024.45  Batch12_2024 27-Jun-24    C-S-3        CS         0
## Batch12_2024.46  Batch12_2024 27-Jun-24    C-S-3        CS        21
## Batch12_2024.47  Batch12_2024 27-Jun-24    C-S-3        CS        42
## Batch12_2024.48  Batch12_2024 27-Jun-24    C-S-3        CS        63
## Batch12_2024.49  Batch12_2024 27-Jun-24    S-M-3        SM         0
```

```
## Batch12_2024.50  Batch12_2024 27-Jun-24    S-M-3        SM        21
## Batch12_2024.51  Batch12_2024 27-Jun-24    S-M-3        SM        42
## Batch12_2024.52  Batch12_2024 27-Jun-24    S-M-3        SM        63
## Batch12_2024.53  Batch12_2024 27-Jun-24  S-M-W-3       SMW         0
## Batch12_2024.54  Batch12_2024 27-Jun-24  S-M-W-3       SMW        21
## Batch12_2024.55  Batch12_2024 27-Jun-24  S-M-W-3       SMW        42
## Batch12_2024.56  Batch12_2024 27-Jun-24  S-M-W-3       SMW        63
## Batch12_2024.57  Batch12_2024 27-Jun-24 C-S-MD-3      CSMD         0
## Batch12_2024.58  Batch12_2024 27-Jun-24 C-S-MD-3      CSMD        21
## Batch12_2024.59  Batch12_2024 27-Jun-24 C-S-MD-3      CSMD        42
## Batch12_2024.60  Batch12_2024 27-Jun-24 C-S-MD-3      CSMD        63
## Batch12_2024.61  Batch12_2024 27-Jun-24 C-S-MD-4      CSMD         0
## Batch12_2024.62  Batch12_2024 27-Jun-24 C-S-MD-4      CSMD        21
## Batch12_2024.63  Batch12_2024 27-Jun-24 C-S-MD-4      CSMD        42
## Batch12_2024.64  Batch12_2024 27-Jun-24 C-S-MD-4      CSMD        63
## Batch12_2024.65  Batch12_2024 27-Jun-24      C-4         C         0
## Batch12_2024.66  Batch12_2024 27-Jun-24      C-4         C        21
## Batch12_2024.67  Batch12_2024 27-Jun-24      C-4         C        42
## Batch12_2024.68  Batch12_2024 27-Jun-24      C-4         C        63
## Batch12_2024.69  Batch12_2024 27-Jun-24  S-M-W-4       SMW         0
## Batch12_2024.70  Batch12_2024 27-Jun-24  S-M-W-4       SMW        21
## Batch12_2024.71  Batch12_2024 27-Jun-24  S-M-W-4       SMW        42
## Batch12_2024.72  Batch12_2024 27-Jun-24  S-M-W-4       SMW        63
## Batch12_2024.73  Batch12_2024 27-Jun-24    S-M-4        SM         0
## Batch12_2024.74  Batch12_2024 27-Jun-24    S-M-4        SM        21
## Batch12_2024.75  Batch12_2024 27-Jun-24    S-M-4        SM        42
## Batch12_2024.76  Batch12_2024 27-Jun-24    S-M-4        SM        63
## Batch12_2024.77  Batch12_2024 27-Jun-24    C-S-4        CS         0
## Batch12_2024.78  Batch12_2024 27-Jun-24    C-S-4        CS        21
## Batch12_2024.79  Batch12_2024 27-Jun-24    C-S-4        CS        42
## Batch12_2024.80  Batch12_2024 27-Jun-24    C-S-4        CS        63
## Batch14_2024.81  Batch14_2024 01-Aug-24  S-M-W-1       SMW         0
## Batch14_2024.82  Batch14_2024 01-Aug-24  S-M-W-1       SMW        21
## Batch14_2024.83  Batch14_2024 01-Aug-24  S-M-W-1       SMW        42
## Batch14_2024.84  Batch14_2024 01-Aug-24  S-M-W-1       SMW        63
## Batch14_2024.85  Batch14_2024 01-Aug-24    C-S-1        CS         0
## Batch14_2024.86  Batch14_2024 01-Aug-24    C-S-1        CS        21
## Batch14_2024.87  Batch14_2024 01-Aug-24    C-S-1        CS        42
## Batch14_2024.88  Batch14_2024 01-Aug-24    C-S-1        CS        63
## Batch14_2024.89  Batch14_2024 01-Aug-24    S-M-1        SM         0
## Batch14_2024.90  Batch14_2024 01-Aug-24    S-M-1        SM        21
## Batch14_2024.91  Batch14_2024 01-Aug-24    S-M-1        SM        42
## Batch14_2024.92  Batch14_2024 01-Aug-24    S-M-1        SM        63
## Batch14_2024.93  Batch14_2024 01-Aug-24 C-S-MD-1      CSMD         0
## Batch14_2024.94  Batch14_2024 01-Aug-24 C-S-MD-1      CSMD        21
## Batch14_2024.95  Batch14_2024 01-Aug-24 C-S-MD-1      CSMD        42
## Batch14_2024.96  Batch14_2024 01-Aug-24 C-S-MD-1      CSMD        63
## Batch14_2024.97  Batch14_2024 01-Aug-24      C-1         C         0
## Batch14_2024.98  Batch14_2024 01-Aug-24      C-1         C        21
## Batch14_2024.99  Batch14_2024 01-Aug-24      C-1         C        42
## Batch14_2024.100 Batch14_2024 01-Aug-24      C-1         C        63
## Batch14_2024.101 Batch14_2024 01-Aug-24 C-S-MD-2      CSMD         0
## Batch14_2024.102 Batch14_2024 01-Aug-24 C-S-MD-2      CSMD        21
## Batch14_2024.103 Batch14_2024 01-Aug-24 C-S-MD-2      CSMD        42
```

```
## Batch14_2024.104 Batch14_2024 01-Aug-24 C-S-MD-2      CSMD        63
## Batch14_2024.105 Batch14_2024 01-Aug-24      C-2         C         0
## Batch14_2024.106 Batch14_2024 01-Aug-24      C-2         C        21
## Batch14_2024.107 Batch14_2024 01-Aug-24      C-2         C        42
## Batch14_2024.108 Batch14_2024 01-Aug-24      C-2         C        63
## Batch14_2024.109 Batch14_2024 01-Aug-24    C-S-2        CS         0
## Batch14_2024.110 Batch14_2024 01-Aug-24    C-S-2        CS        21
## Batch14_2024.111 Batch14_2024 01-Aug-24    C-S-2        CS        42
## Batch14_2024.112 Batch14_2024 01-Aug-24    C-S-2        CS        63
## Batch14_2024.113 Batch14_2024 01-Aug-24    S-M-2        SM         0
## Batch14_2024.114 Batch14_2024 01-Aug-24    S-M-2        SM        21
## Batch14_2024.115 Batch14_2024 01-Aug-24    S-M-2        SM        42
## Batch14_2024.116 Batch14_2024 01-Aug-24    S-M-2        SM        63
## Batch14_2024.117 Batch14_2024 01-Aug-24  S-M-W-2       SMW         0
## Batch14_2024.118 Batch14_2024 01-Aug-24  S-M-W-2       SMW        21
## Batch14_2024.119 Batch14_2024 01-Aug-24  S-M-W-2       SMW        42
## Batch14_2024.120 Batch14_2024 01-Aug-24  S-M-W-2       SMW        63
## Batch14_2024.121 Batch14_2024 01-Aug-24      C-3         C         0
## Batch14_2024.122 Batch14_2024 01-Aug-24      C-3         C        21
## Batch14_2024.123 Batch14_2024 01-Aug-24      C-3         C        42
## Batch14_2024.124 Batch14_2024 01-Aug-24      C-3         C        63
## Batch14_2024.125 Batch14_2024 01-Aug-24    C-S-3        CS         0
## Batch14_2024.126 Batch14_2024 01-Aug-24    C-S-3        CS        21
## Batch14_2024.127 Batch14_2024 01-Aug-24    C-S-3        CS        42
## Batch14_2024.128 Batch14_2024 01-Aug-24    C-S-3        CS        63
## Batch14_2024.129 Batch14_2024 01-Aug-24    S-M-3        SM         0
## Batch14_2024.130 Batch14_2024 01-Aug-24    S-M-3        SM        21
## Batch14_2024.131 Batch14_2024 01-Aug-24    S-M-3        SM        42
## Batch14_2024.132 Batch14_2024 01-Aug-24    S-M-3        SM        63
## Batch14_2024.133 Batch14_2024 01-Aug-24  S-M-W-3       SMW         0
## Batch14_2024.134 Batch14_2024 01-Aug-24  S-M-W-3       SMW        21
## Batch14_2024.135 Batch14_2024 01-Aug-24  S-M-W-3       SMW        42
## Batch14_2024.136 Batch14_2024 01-Aug-24  S-M-W-3       SMW        63
## Batch14_2024.137 Batch14_2024 01-Aug-24 C-S-MD-3      CSMD         0
## Batch14_2024.138 Batch14_2024 01-Aug-24 C-S-MD-3      CSMD        21
## Batch14_2024.139 Batch14_2024 01-Aug-24 C-S-MD-3      CSMD        42
## Batch14_2024.140 Batch14_2024 01-Aug-24 C-S-MD-3      CSMD        63
## Batch14_2024.141 Batch14_2024 01-Aug-24 C-S-MD-4      CSMD         0
## Batch14_2024.142 Batch14_2024 01-Aug-24 C-S-MD-4      CSMD        21
## Batch14_2024.143 Batch14_2024 01-Aug-24 C-S-MD-4      CSMD        42
## Batch14_2024.144 Batch14_2024 01-Aug-24 C-S-MD-4      CSMD        63
## Batch14_2024.145 Batch14_2024 01-Aug-24      C-4         C         0
## Batch14_2024.146 Batch14_2024 01-Aug-24      C-4         C        21
## Batch14_2024.147 Batch14_2024 01-Aug-24      C-4         C        42
## Batch14_2024.148 Batch14_2024 01-Aug-24      C-4         C        63
## Batch14_2024.149 Batch14_2024 01-Aug-24  S-M-W-4       SMW         0
## Batch14_2024.150 Batch14_2024 01-Aug-24  S-M-W-4       SMW        21
## Batch14_2024.151 Batch14_2024 01-Aug-24  S-M-W-4       SMW        42
## Batch14_2024.152 Batch14_2024 01-Aug-24  S-M-W-4       SMW        63
## Batch14_2024.153 Batch14_2024 01-Aug-24    S-M-4        SM         0
## Batch14_2024.154 Batch14_2024 01-Aug-24    S-M-4        SM        21
## Batch14_2024.155 Batch14_2024 01-Aug-24    S-M-4        SM        42
## Batch14_2024.156 Batch14_2024 01-Aug-24    S-M-4        SM        63
## Batch14_2024.157 Batch14_2024 01-Aug-24    C-S-4        CS         0
```

```
## Batch14_2024.158 Batch14_2024 01-Aug-24    C-S-4        CS        21
## Batch14_2024.159 Batch14_2024 01-Aug-24    C-S-4        CS        42
## Batch14_2024.160 Batch14_2024 01-Aug-24    C-S-4        CS        63
##                  Chamber_Temp_C Surface_Area_m2 Height_m Volume_m3
## Batch12_2024.1             32.3      0.06834928  0.32475        NA
## Batch12_2024.2             31.7      0.06834928  0.32475        NA
## Batch12_2024.3             33.0      0.06834928  0.32475        NA
## Batch12_2024.4             32.5      0.06834928  0.32475        NA
## Batch12_2024.5             33.7      0.06834928  0.32200        NA
## Batch12_2024.6             34.7      0.06834928  0.32200        NA
## Batch12_2024.7             34.8      0.06834928  0.32200        NA
## Batch12_2024.8             34.5      0.06834928  0.32200        NA
## Batch12_2024.9             33.7      0.06834928  0.30600        NA
## Batch12_2024.10            33.4      0.06834928  0.30600        NA
## Batch12_2024.11            33.2      0.06834928  0.30600        NA
## Batch12_2024.12            33.0      0.06834928  0.30600        NA
## Batch12_2024.13            31.6      0.06834928  0.31100        NA
## Batch12_2024.14            32.1      0.06834928  0.31100        NA
## Batch12_2024.15            31.8      0.06834928  0.31100        NA
## Batch12_2024.16            31.6      0.06834928  0.31100        NA
## Batch12_2024.17            33.0      0.06834928  0.29600        NA
## Batch12_2024.18            33.0      0.06834928  0.29600        NA
## Batch12_2024.19            33.3      0.06834928  0.29600        NA
## Batch12_2024.20            33.0      0.06834928  0.29600        NA
## Batch12_2024.21            30.0      0.06834928  0.31000        NA
## Batch12_2024.22            29.9      0.06834928  0.31000        NA
## Batch12_2024.23            29.7      0.06834928  0.31000        NA
## Batch12_2024.24            29.3      0.06834928  0.31000        NA
## Batch12_2024.25            32.1      0.06834928  0.29800        NA
## Batch12_2024.26            30.9      0.06834928  0.29800        NA
## Batch12_2024.27            31.8      0.06834928  0.29800        NA
## Batch12_2024.28            31.9      0.06834928  0.29800        NA
## Batch12_2024.29            32.1      0.06834928  0.30500        NA
## Batch12_2024.30            30.4      0.06834928  0.30500        NA
## Batch12_2024.31            32.5      0.06834928  0.30500        NA
## Batch12_2024.32            32.7      0.06834928  0.30500        NA
## Batch12_2024.33            31.2      0.06834928  0.30650        NA
## Batch12_2024.34            29.2      0.06834928  0.30650        NA
## Batch12_2024.35            31.0      0.06834928  0.30650        NA
## Batch12_2024.36            29.8      0.06834928  0.30650        NA
## Batch12_2024.37            30.4      0.06834928  0.27400        NA
## Batch12_2024.38            29.8      0.06834928  0.27400        NA
## Batch12_2024.39            31.8      0.06834928  0.27400        NA
## Batch12_2024.40            32.1      0.06834928  0.27400        NA
## Batch12_2024.41            29.1      0.06834928  0.30650        NA
## Batch12_2024.42            31.3      0.06834928  0.30650        NA
## Batch12_2024.43            32.2      0.06834928  0.30650        NA
## Batch12_2024.44            31.9      0.06834928  0.30650        NA
## Batch12_2024.45            38.6      0.06834928  0.30550        NA
## Batch12_2024.46            37.1      0.06834928  0.30550        NA
## Batch12_2024.47            40.0      0.06834928  0.30550        NA
## Batch12_2024.48            33.7      0.06834928  0.30550        NA
## Batch12_2024.49            31.1      0.06834928  0.30300        NA
## Batch12_2024.50            32.4      0.06834928  0.30300        NA
```

```
## Batch12_2024.51            32.6      0.06834928  0.30300         NA
## Batch12_2024.52            32.4      0.06834928  0.30300         NA
## Batch12_2024.53            30.4      0.06834928  0.30100         NA
## Batch12_2024.54            32.6      0.06834928  0.30100         NA
## Batch12_2024.55            32.5      0.06834928  0.30100         NA
## Batch12_2024.56            31.4      0.06834928  0.30100         NA
## Batch12_2024.57            31.9      0.06834928  0.30750         NA
## Batch12_2024.58            32.7      0.06834928  0.30750         NA
## Batch12_2024.59            32.9      0.06834928  0.30750         NA
## Batch12_2024.60            32.9      0.06834928  0.30750         NA
## Batch12_2024.61            29.4      0.06834928  0.31050         NA
## Batch12_2024.62            29.7      0.06834928  0.31050         NA
## Batch12_2024.63            29.7      0.06834928  0.31050         NA
## Batch12_2024.64            29.9      0.06834928  0.31050         NA
## Batch12_2024.65            27.6      0.06834928  0.30225         NA
## Batch12_2024.66            30.3      0.06834928  0.30225         NA
## Batch12_2024.67            30.3      0.06834928  0.30225         NA
## Batch12_2024.68            29.4      0.06834928  0.30225         NA
## Batch12_2024.69            28.4      0.06834928  0.30050         NA
## Batch12_2024.70            30.5      0.06834928  0.30050         NA
## Batch12_2024.71            34.2      0.06834928  0.30050         NA
## Batch12_2024.72            31.5      0.06834928  0.30050         NA
## Batch12_2024.73            27.8      0.06834928  0.30150         NA
## Batch12_2024.74            31.5      0.06834928  0.30150         NA
## Batch12_2024.75            31.6      0.06834928  0.30150         NA
## Batch12_2024.76            30.1      0.06834928  0.30150         NA
## Batch12_2024.77            29.6      0.06834928  0.30150         NA
## Batch12_2024.78            30.6      0.06834928  0.30150         NA
## Batch12_2024.79            32.0      0.06834928  0.30150         NA
## Batch12_2024.80            30.6      0.06834928  0.30150         NA
## Batch14_2024.81            35.2      0.06834928  0.81070         NA
## Batch14_2024.82            36.3      0.06834928  0.81070         NA
## Batch14_2024.83            36.9      0.06834928  0.81070         NA
## Batch14_2024.84            37.8      0.06834928  0.81070         NA
## Batch14_2024.85            33.9      0.06834928  0.80870         NA
## Batch14_2024.86            36.3      0.06834928  0.80870         NA
## Batch14_2024.87            37.9      0.06834928  0.80870         NA
## Batch14_2024.88            38.2      0.06834928  0.80870         NA
## Batch14_2024.89            36.0      0.06834928  0.79570         NA
## Batch14_2024.90            34.4      0.06834928  0.79570         NA
## Batch14_2024.91            35.9      0.06834928  0.79570         NA
## Batch14_2024.92            35.0      0.06834928  0.79570         NA
## Batch14_2024.93            36.0      0.06834928  0.79645         NA
## Batch14_2024.94            36.6      0.06834928  0.79645         NA
## Batch14_2024.95            39.1      0.06834928  0.79645         NA
## Batch14_2024.96            39.1      0.06834928  0.79645         NA
## Batch14_2024.97            39.9      0.06834928  0.78145         NA
## Batch14_2024.98            40.9      0.06834928  0.78145         NA
## Batch14_2024.99            42.9      0.06834928  0.78145         NA
## Batch14_2024.100           42.7      0.06834928  0.78145         NA
## Batch14_2024.101           34.6      0.06834928  0.79495         NA
## Batch14_2024.102           36.1      0.06834928  0.79495         NA
## Batch14_2024.103           36.3      0.06834928  0.79495         NA
## Batch14_2024.104           37.8      0.06834928  0.79495         NA
```

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## Batch14_2024.105           36.4      0.06834928  0.78420         NA
## Batch14_2024.106           38.9      0.06834928  0.78420         NA
## Batch14_2024.107           39.7      0.06834928  0.78420         NA
## Batch14_2024.108           41.0      0.06834928  0.78420         NA
## Batch14_2024.109           35.6      0.06834928  0.79020         NA
## Batch14_2024.110           37.7      0.06834928  0.79020         NA
## Batch14_2024.111           37.5      0.06834928  0.79020         NA
## Batch14_2024.112           38.1      0.06834928  0.79020         NA
## Batch14_2024.113           36.8      0.06834928  0.79695         NA
## Batch14_2024.114           37.4      0.06834928  0.79695         NA
## Batch14_2024.115           38.3      0.06834928  0.79695         NA
## Batch14_2024.116           38.3      0.06834928  0.79695         NA
## Batch14_2024.117           40.8      0.06834928  0.75945         NA
## Batch14_2024.118           42.0      0.06834928  0.75945         NA
## Batch14_2024.119           41.7      0.06834928  0.75945         NA
## Batch14_2024.120           43.1      0.06834928  0.75945         NA
## Batch14_2024.121           37.5      0.06834928  0.79595         NA
## Batch14_2024.122           40.7      0.06834928  0.79595         NA
## Batch14_2024.123           40.8      0.06834928  0.79595         NA
## Batch14_2024.124           42.4      0.06834928  0.79595         NA
## Batch14_2024.125           37.1      0.06834928  0.79570         NA
## Batch14_2024.126           37.3      0.06834928  0.79570         NA
## Batch14_2024.127           37.9      0.06834928  0.79570         NA
## Batch14_2024.128           39.5      0.06834928  0.79570         NA
## Batch14_2024.129           33.6      0.06834928  0.79270         NA
## Batch14_2024.130           37.8      0.06834928  0.79270         NA
## Batch14_2024.131           38.1      0.06834928  0.79270         NA
## Batch14_2024.132           39.4      0.06834928  0.79270         NA
## Batch14_2024.133           39.1      0.06834928  0.78895         NA
## Batch14_2024.134           39.9      0.06834928  0.78895         NA
## Batch14_2024.135           41.2      0.06834928  0.78895         NA
## Batch14_2024.136           41.5      0.06834928  0.78895         NA
## Batch14_2024.137           36.1      0.06834928  0.79770         NA
## Batch14_2024.138           38.1      0.06834928  0.79770         NA
## Batch14_2024.139           40.1      0.06834928  0.79770         NA
## Batch14_2024.140           41.0      0.06834928  0.79770         NA
## Batch14_2024.141           34.9      0.06834928  0.79970         NA
## Batch14_2024.142           35.6      0.06834928  0.79970         NA
## Batch14_2024.143           38.2      0.06834928  0.79970         NA
## Batch14_2024.144           41.2      0.06834928  0.79970         NA
## Batch14_2024.145           38.9      0.06834928  0.78970         NA
## Batch14_2024.146           37.8      0.06834928  0.78970         NA
## Batch14_2024.147           39.0      0.06834928  0.78970         NA
## Batch14_2024.148           37.6      0.06834928  0.78970         NA
## Batch14_2024.149           36.2      0.06834928  0.78995         NA
## Batch14_2024.150           36.6      0.06834928  0.78995         NA
## Batch14_2024.151           36.8      0.06834928  0.78995         NA
## Batch14_2024.152           37.5      0.06834928  0.78995         NA
## Batch14_2024.153           40.0      0.06834928  0.79270         NA
## Batch14_2024.154           41.5      0.06834928  0.79270         NA
## Batch14_2024.155           42.8      0.06834928  0.79270         NA
## Batch14_2024.156           43.1      0.06834928  0.79270         NA
## Batch14_2024.157           36.3      0.06834928  0.79070         NA
## Batch14_2024.158           41.3      0.06834928  0.79070         NA
```

```
## Batch14_2024.159          42.7      0.06834928  0.79070        NA
## Batch14_2024.160          42.9      0.06834928  0.79070        NA
##                    Sample_CH4_Peak Sample_CH4_ppm Sample_CO2_Peak Sample_CO2_ppm
## Batch12_2024.1               59701      10.860764              NA             NA
## Batch12_2024.2              144169      25.988029              NA             NA
## Batch12_2024.3              227122      40.843975              NA             NA
## Batch12_2024.4              310759      55.822417              NA             NA
## Batch12_2024.5               45013       8.230309              NA             NA
## Batch12_2024.6               82035      14.860531              NA             NA
## Batch12_2024.7              114774      20.723716              NA             NA
## Batch12_2024.8              144773      26.096199              NA             NA
## Batch12_2024.9               51940       9.470856              NA             NA
## Batch12_2024.10              84241      15.255601              NA             NA
## Batch12_2024.11             112839      20.377180              NA             NA
## Batch12_2024.12             137742      24.837026              NA             NA
## Batch12_2024.13              70226      12.745673              NA             NA
## Batch12_2024.14             118334      21.361272              NA             NA
## Batch12_2024.15             159713      28.771784              NA             NA
## Batch12_2024.16             207247      37.284587              NA             NA
## Batch12_2024.17             259633      46.666328              NA             NA
## Batch12_2024.18             410334      73.655176              NA             NA
## Batch12_2024.19             573167     102.816727              NA             NA
## Batch12_2024.20             746536     133.865158              NA             NA
## Batch12_2024.21              17655       3.330800              NA             NA
## Batch12_2024.22              50011       9.125394              NA             NA
## Batch12_2024.23              78440      14.216707              NA             NA
## Batch12_2024.24             106280      19.202537              NA             NA
## Batch12_2024.25              16449       3.114819              NA             NA
## Batch12_2024.26             163679      29.482050              NA             NA
## Batch12_2024.27             306079      54.984282              NA             NA
## Batch12_2024.28             442606      79.434726              NA             NA
## Batch12_2024.29              14666       2.795504              NA             NA
## Batch12_2024.30              37138       6.819985              NA             NA
## Batch12_2024.31              53796       9.803245              NA             NA
## Batch12_2024.32              75401      13.672457              NA             NA
## Batch12_2024.33              13250       2.541914              NA             NA
## Batch12_2024.34              64316      11.687258              NA             NA
## Batch12_2024.35             112706      20.353361              NA             NA
## Batch12_2024.36             159532      28.739369              NA             NA
## Batch12_2024.37              15430       2.932328              NA             NA
## Batch12_2024.38             139978      25.237468              NA             NA
## Batch12_2024.39             260619      46.842909              NA             NA
## Batch12_2024.40             386882      69.455187              NA             NA
## Batch12_2024.41              93759      16.960167              NA             NA
## Batch12_2024.42             295171      53.030782              NA             NA
## Batch12_2024.43             482628      86.602215              NA             NA
## Batch12_2024.44             657941     117.998794              NA             NA
## Batch12_2024.45              83602      15.141163              NA             NA
## Batch12_2024.46             175826      31.657440              NA             NA
## Batch12_2024.47             263411      47.342925              NA             NA
## Batch12_2024.48             343467      61.680051              NA             NA
## Batch12_2024.49              39227       7.194102              NA             NA
## Batch12_2024.50              64061      11.641591              NA             NA
## Batch12_2024.51              87789      15.891008              NA             NA
```

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## Batch12_2024.52          110621      19.979961            NA            NA
## Batch12_2024.53           79580      14.420868            NA            NA
## Batch12_2024.54          132422      23.884274            NA            NA
## Batch12_2024.55          193813      34.878709            NA            NA
## Batch12_2024.56          247312      44.459776            NA            NA
## Batch12_2024.57           85507      15.482327            NA            NA
## Batch12_2024.58          134387      24.236183            NA            NA
## Batch12_2024.59          189081      34.031261            NA            NA
## Batch12_2024.60          233531      41.991754            NA            NA
## Batch12_2024.61           19762       3.708140            NA            NA
## Batch12_2024.62           68369      12.413105            NA            NA
## Batch12_2024.63          110988      20.045687            NA            NA
## Batch12_2024.64          151773      27.349820            NA            NA
## Batch12_2024.65           16788       3.175530            NA            NA
## Batch12_2024.66          133189      24.021635            NA            NA
## Batch12_2024.67          241222      43.369126            NA            NA
## Batch12_2024.68          351103      63.047572            NA            NA
## Batch12_2024.69           15137       2.879855            NA            NA
## Batch12_2024.70           58962      10.728417            NA            NA
## Batch12_2024.71          100696      18.202506            NA            NA
## Batch12_2024.72          142020      25.603167            NA            NA
## Batch12_2024.73           14840       2.826665            NA            NA
## Batch12_2024.74           59950      10.905357            NA            NA
## Batch12_2024.75          103659      18.733145            NA            NA
## Batch12_2024.76          145397      26.207950            NA            NA
## Batch12_2024.77           13342       2.558390            NA            NA
## Batch12_2024.78           51088       9.318273            NA            NA
## Batch12_2024.79           86432      15.647984            NA            NA
## Batch12_2024.80          119801      21.623995            NA            NA
## Batch14_2024.81           84743      15.034195            NA            NA
## Batch14_2024.82          236721      41.747879            NA            NA
## Batch14_2024.83          426998      75.193507            NA            NA
## Batch14_2024.84          604575     106.406813            NA            NA
## Batch14_2024.85           64968      11.558277            NA            NA
## Batch14_2024.86          190049      33.544185            NA            NA
## Batch14_2024.87          343349      60.490241            NA            NA
## Batch14_2024.88          498803      87.814913            NA            NA
## Batch14_2024.89           93336      16.544616            NA            NA
## Batch14_2024.90          229867      40.543128            NA            NA
## Batch14_2024.91          382312      67.338898            NA            NA
## Batch14_2024.92          521456      91.796707            NA            NA
## Batch14_2024.93           61272      10.908619            NA            NA
## Batch14_2024.94          208670      36.817261            NA            NA
## Batch14_2024.95          382880      67.438738            NA            NA
## Batch14_2024.96          550141      96.838766            NA            NA
## Batch14_2024.97           62594      11.140991            NA            NA
## Batch14_2024.98          205119      36.193089            NA            NA
## Batch14_2024.99          348819      61.451721            NA            NA
## Batch14_2024.100         475145      83.656467            NA            NA
## Batch14_2024.101          48106       8.594387            NA            NA
## Batch14_2024.102         149608      26.435734            NA            NA
## Batch14_2024.103         270450      47.676539            NA            NA
## Batch14_2024.104         385629      67.921939            NA            NA
## Batch14_2024.105          58467      10.415575            NA            NA
```

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## Batch14_2024.106          174309      30.777512              NA              NA
## Batch14_2024.107          310583      54.730851              NA              NA
## Batch14_2024.108          437127      76.973915              NA              NA
## Batch14_2024.109           66194      11.773775              NA              NA
## Batch14_2024.110          158631      28.021737              NA              NA
## Batch14_2024.111          259664      45.780647              NA              NA
## Batch14_2024.112          352226      62.050581              NA              NA
## Batch14_2024.113          104024      18.423282              NA              NA
## Batch14_2024.114          275683      48.596361              NA              NA
## Batch14_2024.115          458974      80.814036              NA              NA
## Batch14_2024.116          632065     111.238822              NA              NA
## Batch14_2024.117          107580      19.048332              NA              NA
## Batch14_2024.118          309473      54.535743              NA              NA
## Batch14_2024.119          505344      88.964646              NA              NA
## Batch14_2024.120          756101     133.041047              NA              NA
## Batch14_2024.121           48004       8.576458              NA              NA
## Batch14_2024.122          132319      23.396789              NA              NA
## Batch14_2024.123          294552      51.913028              NA              NA
## Batch14_2024.124          389539      68.609212              NA              NA
## Batch14_2024.125           59909      10.669040              NA              NA
## Batch14_2024.126          155927      27.546446              NA              NA
## Batch14_2024.127          315294      55.558919              NA              NA
## Batch14_2024.128          510359      89.846150              NA              NA
## Batch14_2024.129           68567      12.190886              NA              NA
## Batch14_2024.130          144589      25.553528              NA              NA
## Batch14_2024.131          269624      47.531350              NA              NA
## Batch14_2024.132          339633      59.837067              NA              NA
## Batch14_2024.133           90550      16.054911              NA              NA
## Batch14_2024.134          206925      36.510536              NA              NA
## Batch14_2024.135          386925      68.149741              NA              NA
## Batch14_2024.136          538881      94.859557              NA              NA
## Batch14_2024.137           95497      16.924462              NA              NA
## Batch14_2024.138          176183      31.106911              NA              NA
## Batch14_2024.139          342502      60.341361              NA              NA
## Batch14_2024.140          435742      76.730469              NA              NA
## Batch14_2024.141           46697       8.346722              NA              NA
## Batch14_2024.142          137366      24.283917              NA              NA
## Batch14_2024.143          255705      45.084761              NA              NA
## Batch14_2024.144          363809      64.086564              NA              NA
## Batch14_2024.145           44423       7.947014              NA              NA
## Batch14_2024.146          131822      23.309429              NA              NA
## Batch14_2024.147          260479      45.923903              NA              NA
## Batch14_2024.148          359000      63.241270              NA              NA
## Batch14_2024.149           87014      15.433377              NA              NA
## Batch14_2024.150          208288      36.750115              NA              NA
## Batch14_2024.151          315120      55.528335              NA              NA
## Batch14_2024.152          501953      88.368599              NA              NA
## Batch14_2024.153           67372      11.980836              NA              NA
## Batch14_2024.154          191454      33.791146              NA              NA
## Batch14_2024.155          326728      57.568712              NA              NA
## Batch14_2024.156          464688      81.818405              NA              NA
## Batch14_2024.157           79001      14.024905              NA              NA
## Batch14_2024.158          181124      31.975408              NA              NA
## Batch14_2024.159          294139      51.840434              NA              NA
```

```
## Batch14_2024.160          396874      69.898510               NA              NA
##                    Sample_N2O_Peak Sample_N2O_ppm Sample_Gas1_Peak
## Batch12_2024.1               40962      0.3164698               NA
## Batch12_2024.2               40955      0.3163928               NA
## Batch12_2024.3               40341      0.3096409               NA
## Batch12_2024.4               41718      0.3247831               NA
## Batch12_2024.5               40518      0.3115873               NA
## Batch12_2024.6               39510      0.3005028               NA
## Batch12_2024.7               40220      0.3083103               NA
## Batch12_2024.8               40269      0.3088492               NA
## Batch12_2024.9               41504      0.3224299               NA
## Batch12_2024.10              40712      0.3137206               NA
## Batch12_2024.11              42109      0.3290828               NA
## Batch12_2024.12              40429      0.3106086               NA
## Batch12_2024.13              41026      0.3171735               NA
## Batch12_2024.14              40463      0.3109825               NA
## Batch12_2024.15              40304      0.3092341               NA
## Batch12_2024.16              41072      0.3176794               NA
## Batch12_2024.17              40182      0.3078925               NA
## Batch12_2024.18              41175      0.3188120               NA
## Batch12_2024.19              41496      0.3223419               NA
## Batch12_2024.20              41173      0.3187900               NA
## Batch12_2024.21              41527      0.3226828               NA
## Batch12_2024.22              41165      0.3187021               NA
## Batch12_2024.23              41749      0.3251240               NA
## Batch12_2024.24              41118      0.3181852               NA
## Batch12_2024.25              41487      0.3222429               NA
## Batch12_2024.26              41687      0.3244422               NA
## Batch12_2024.27              40372      0.3099818               NA
## Batch12_2024.28              41408      0.3213742               NA
## Batch12_2024.29              39280      0.2979736               NA
## Batch12_2024.30              42572      0.3341741               NA
## Batch12_2024.31              41743      0.3250580               NA
## Batch12_2024.32              41325      0.3204615               NA
## Batch12_2024.33              42515      0.3335473               NA
## Batch12_2024.34              41613      0.3236285               NA
## Batch12_2024.35              41854      0.3262786               NA
## Batch12_2024.36              41756      0.3252010               NA
## Batch12_2024.37              41572      0.3231776               NA
## Batch12_2024.38              40594      0.3124230               NA
## Batch12_2024.39              40583      0.3123021               NA
## Batch12_2024.40              42418      0.3324807               NA
## Batch12_2024.41              42198      0.3300614               NA
## Batch12_2024.42              42378      0.3320408               NA
## Batch12_2024.43              41735      0.3249701               NA
## Batch12_2024.44              40867      0.3154251               NA
## Batch12_2024.45              40434      0.3106636               NA
## Batch12_2024.46              42090      0.3288738               NA
## Batch12_2024.47              42074      0.3286979               NA
## Batch12_2024.48              40573      0.3121921               NA
## Batch12_2024.49              39929      0.3051104               NA
## Batch12_2024.50              41496      0.3223419               NA
## Batch12_2024.51              41444      0.3217701               NA
## Batch12_2024.52              40685      0.3134237               NA
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## Batch12_2024.53              40699      0.3135777               NA
## Batch12_2024.54              39834      0.3040657               NA
## Batch12_2024.55              40010      0.3060011               NA
## Batch12_2024.56              40728      0.3138966               NA
## Batch12_2024.57              40651      0.3130498               NA
## Batch12_2024.58              41069      0.3176464               NA
## Batch12_2024.59              40900      0.3157880               NA
## Batch12_2024.60              40733      0.3139516               NA
## Batch12_2024.61              41891      0.3266855               NA
## Batch12_2024.62              40459      0.3109385               NA
## Batch12_2024.63              40833      0.3150512               NA
## Batch12_2024.64              40002      0.3059131               NA
## Batch12_2024.65              41711      0.3247061               NA
## Batch12_2024.66              41242      0.3195488               NA
## Batch12_2024.67              41370      0.3209563               NA
## Batch12_2024.68              42500      0.3333824               NA
## Batch12_2024.69              41343      0.3206594               NA
## Batch12_2024.70              40461      0.3109605               NA
## Batch12_2024.71              39589      0.3013715               NA
## Batch12_2024.72              40661      0.3131598               NA
## Batch12_2024.73              40029      0.3062100               NA
## Batch12_2024.74              40432      0.3106416               NA
## Batch12_2024.75              40901      0.3157990               NA
## Batch12_2024.76              39303      0.2982265               NA
## Batch12_2024.77              40850      0.3152382               NA
## Batch12_2024.78              41482      0.3221879               NA
## Batch12_2024.79              40181      0.3078815               NA
## Batch12_2024.80              41238      0.3195048               NA
## Batch14_2024.81              36840      0.3088018               NA
## Batch14_2024.82              38041      0.3226310               NA
## Batch14_2024.83              39207      0.3360571               NA
## Batch14_2024.84              38374      0.3264654               NA
## Batch14_2024.85              38120      0.3235407               NA
## Batch14_2024.86              37779      0.3196141               NA
## Batch14_2024.87              38058      0.3228267               NA
## Batch14_2024.88              39641      0.3410545               NA
## Batch14_2024.89              37371      0.3149161               NA
## Batch14_2024.90              37933      0.3213874               NA
## Batch14_2024.91              38151      0.3238976               NA
## Batch14_2024.92              38473      0.3276053               NA
## Batch14_2024.93              38474      0.3276169               NA
## Batch14_2024.94              38026      0.3224583               NA
## Batch14_2024.95              38136      0.3237249               NA
## Batch14_2024.96              38773      0.3310598               NA
## Batch14_2024.97              38683      0.3300234               NA
## Batch14_2024.98              39025      0.3339615               NA
## Batch14_2024.99              38302      0.3256363               NA
## Batch14_2024.100             38061      0.3228613               NA
## Batch14_2024.101             38521      0.3281581               NA
## Batch14_2024.102             38252      0.3250606               NA
## Batch14_2024.103             37997      0.3221243               NA
## Batch14_2024.104             38158      0.3239782               NA
## Batch14_2024.105             38134      0.3237019               NA
## Batch14_2024.106             38979      0.3334318               NA
```

```
## Batch14_2024.107            39183      0.3357808                NA
## Batch14_2024.108            38263      0.3251873                NA
## Batch14_2024.109            37970      0.3218134                NA
## Batch14_2024.110            38147      0.3238515                NA
## Batch14_2024.111            38514      0.3280774                NA
## Batch14_2024.112            37741      0.3191766                NA
## Batch14_2024.113            37052      0.3112429                NA
## Batch14_2024.114            37853      0.3204662                NA
## Batch14_2024.115            38023      0.3224237                NA
## Batch14_2024.116            38007      0.3222395                NA
## Batch14_2024.117            36974      0.3103448                NA
## Batch14_2024.118            37877      0.3207426                NA
## Batch14_2024.119            37250      0.3135228                NA
## Batch14_2024.120            37451      0.3158373                NA
## Batch14_2024.121            38154      0.3239322                NA
## Batch14_2024.122            38007      0.3222395                NA
## Batch14_2024.123            38130      0.3236558                NA
## Batch14_2024.124            37368      0.3148816                NA
## Batch14_2024.125            37739      0.3191535                NA
## Batch14_2024.126            38789      0.3312440                NA
## Batch14_2024.127            38716      0.3304034                NA
## Batch14_2024.128            37729      0.3190384                NA
## Batch14_2024.129            37717      0.3189002                NA
## Batch14_2024.130            37559      0.3170809                NA
## Batch14_2024.131            38711      0.3303458                NA
## Batch14_2024.132            39013      0.3338233                NA
## Batch14_2024.133            39329      0.3374619                NA
## Batch14_2024.134            38148      0.3238631                NA
## Batch14_2024.135            37890      0.3208923                NA
## Batch14_2024.136            38807      0.3314513                NA
## Batch14_2024.137            38083      0.3231146                NA
## Batch14_2024.138            37063      0.3113696                NA
## Batch14_2024.139            38043      0.3226540                NA
## Batch14_2024.140            38273      0.3253024                NA
## Batch14_2024.141            37892      0.3209153                NA
## Batch14_2024.142            39157      0.3354814                NA
## Batch14_2024.143            37276      0.3138222                NA
## Batch14_2024.144            39697      0.3416994                NA
## Batch14_2024.145            38842      0.3318543                NA
## Batch14_2024.146            38210      0.3245770                NA
## Batch14_2024.147            37835      0.3202590                NA
## Batch14_2024.148            38351      0.3262005                NA
## Batch14_2024.149            37008      0.3107363                NA
## Batch14_2024.150            38415      0.3269375                NA
## Batch14_2024.151            37838      0.3202935                NA
## Batch14_2024.152            38001      0.3221704                NA
## Batch14_2024.153            38070      0.3229649                NA
## Batch14_2024.154            38759      0.3308986                NA
## Batch14_2024.155            38090      0.3231952                NA
## Batch14_2024.156            35822      0.2970798                NA
## Batch14_2024.157            37746      0.3192341                NA
## Batch14_2024.158            38090      0.3231952                NA
## Batch14_2024.159            37199      0.3129356                NA
## Batch14_2024.160            38621      0.3293095                NA
```

```
##                   Sample_Gas1_ppm Sample_Gas2_Peak Sample_Gas2_ppm
## Batch12_2024.1                 NA               NA              NA
## Batch12_2024.2                 NA               NA              NA
## Batch12_2024.3                 NA               NA              NA
## Batch12_2024.4                 NA               NA              NA
## Batch12_2024.5                 NA               NA              NA
## Batch12_2024.6                 NA               NA              NA
## Batch12_2024.7                 NA               NA              NA
## Batch12_2024.8                 NA               NA              NA
## Batch12_2024.9                 NA               NA              NA
## Batch12_2024.10                NA               NA              NA
## Batch12_2024.11                NA               NA              NA
## Batch12_2024.12                NA               NA              NA
## Batch12_2024.13                NA               NA              NA
## Batch12_2024.14                NA               NA              NA
## Batch12_2024.15                NA               NA              NA
## Batch12_2024.16                NA               NA              NA
## Batch12_2024.17                NA               NA              NA
## Batch12_2024.18                NA               NA              NA
## Batch12_2024.19                NA               NA              NA
## Batch12_2024.20                NA               NA              NA
## Batch12_2024.21                NA               NA              NA
## Batch12_2024.22                NA               NA              NA
## Batch12_2024.23                NA               NA              NA
## Batch12_2024.24                NA               NA              NA
## Batch12_2024.25                NA               NA              NA
## Batch12_2024.26                NA               NA              NA
## Batch12_2024.27                NA               NA              NA
## Batch12_2024.28                NA               NA              NA
## Batch12_2024.29                NA               NA              NA
## Batch12_2024.30                NA               NA              NA
## Batch12_2024.31                NA               NA              NA
## Batch12_2024.32                NA               NA              NA
## Batch12_2024.33                NA               NA              NA
## Batch12_2024.34                NA               NA              NA
## Batch12_2024.35                NA               NA              NA
## Batch12_2024.36                NA               NA              NA
## Batch12_2024.37                NA               NA              NA
## Batch12_2024.38                NA               NA              NA
## Batch12_2024.39                NA               NA              NA
## Batch12_2024.40                NA               NA              NA
## Batch12_2024.41                NA               NA              NA
## Batch12_2024.42                NA               NA              NA
## Batch12_2024.43                NA               NA              NA
## Batch12_2024.44                NA               NA              NA
## Batch12_2024.45                NA               NA              NA
## Batch12_2024.46                NA               NA              NA
## Batch12_2024.47                NA               NA              NA
## Batch12_2024.48                NA               NA              NA
## Batch12_2024.49                NA               NA              NA
## Batch12_2024.50                NA               NA              NA
## Batch12_2024.51                NA               NA              NA
## Batch12_2024.52                NA               NA              NA
## Batch12_2024.53                NA               NA              NA
```

```
## Batch12_2024.54                NA                NA                NA
## Batch12_2024.55                NA                NA                NA
## Batch12_2024.56                NA                NA                NA
## Batch12_2024.57                NA                NA                NA
## Batch12_2024.58                NA                NA                NA
## Batch12_2024.59                NA                NA                NA
## Batch12_2024.60                NA                NA                NA
## Batch12_2024.61                NA                NA                NA
## Batch12_2024.62                NA                NA                NA
## Batch12_2024.63                NA                NA                NA
## Batch12_2024.64                NA                NA                NA
## Batch12_2024.65                NA                NA                NA
## Batch12_2024.66                NA                NA                NA
## Batch12_2024.67                NA                NA                NA
## Batch12_2024.68                NA                NA                NA
## Batch12_2024.69                NA                NA                NA
## Batch12_2024.70                NA                NA                NA
## Batch12_2024.71                NA                NA                NA
## Batch12_2024.72                NA                NA                NA
## Batch12_2024.73                NA                NA                NA
## Batch12_2024.74                NA                NA                NA
## Batch12_2024.75                NA                NA                NA
## Batch12_2024.76                NA                NA                NA
## Batch12_2024.77                NA                NA                NA
## Batch12_2024.78                NA                NA                NA
## Batch12_2024.79                NA                NA                NA
## Batch12_2024.80                NA                NA                NA
## Batch14_2024.81                NA                NA                NA
## Batch14_2024.82                NA                NA                NA
## Batch14_2024.83                NA                NA                NA
## Batch14_2024.84                NA                NA                NA
## Batch14_2024.85                NA                NA                NA
## Batch14_2024.86                NA                NA                NA
## Batch14_2024.87                NA                NA                NA
## Batch14_2024.88                NA                NA                NA
## Batch14_2024.89                NA                NA                NA
## Batch14_2024.90                NA                NA                NA
## Batch14_2024.91                NA                NA                NA
## Batch14_2024.92                NA                NA                NA
## Batch14_2024.93                NA                NA                NA
## Batch14_2024.94                NA                NA                NA
## Batch14_2024.95                NA                NA                NA
## Batch14_2024.96                NA                NA                NA
## Batch14_2024.97                NA                NA                NA
## Batch14_2024.98                NA                NA                NA
## Batch14_2024.99                NA                NA                NA
## Batch14_2024.100               NA                NA                NA
## Batch14_2024.101               NA                NA                NA
## Batch14_2024.102               NA                NA                NA
## Batch14_2024.103               NA                NA                NA
## Batch14_2024.104               NA                NA                NA
## Batch14_2024.105               NA                NA                NA
## Batch14_2024.106               NA                NA                NA
## Batch14_2024.107               NA                NA                NA
```

```
## Batch14_2024.108              NA              NA              NA
## Batch14_2024.109              NA              NA              NA
## Batch14_2024.110              NA              NA              NA
## Batch14_2024.111              NA              NA              NA
## Batch14_2024.112              NA              NA              NA
## Batch14_2024.113              NA              NA              NA
## Batch14_2024.114              NA              NA              NA
## Batch14_2024.115              NA              NA              NA
## Batch14_2024.116              NA              NA              NA
## Batch14_2024.117              NA              NA              NA
## Batch14_2024.118              NA              NA              NA
## Batch14_2024.119              NA              NA              NA
## Batch14_2024.120              NA              NA              NA
## Batch14_2024.121              NA              NA              NA
## Batch14_2024.122              NA              NA              NA
## Batch14_2024.123              NA              NA              NA
## Batch14_2024.124              NA              NA              NA
## Batch14_2024.125              NA              NA              NA
## Batch14_2024.126              NA              NA              NA
## Batch14_2024.127              NA              NA              NA
## Batch14_2024.128              NA              NA              NA
## Batch14_2024.129              NA              NA              NA
## Batch14_2024.130              NA              NA              NA
## Batch14_2024.131              NA              NA              NA
## Batch14_2024.132              NA              NA              NA
## Batch14_2024.133              NA              NA              NA
## Batch14_2024.134              NA              NA              NA
## Batch14_2024.135              NA              NA              NA
## Batch14_2024.136              NA              NA              NA
## Batch14_2024.137              NA              NA              NA
## Batch14_2024.138              NA              NA              NA
## Batch14_2024.139              NA              NA              NA
## Batch14_2024.140              NA              NA              NA
## Batch14_2024.141              NA              NA              NA
## Batch14_2024.142              NA              NA              NA
## Batch14_2024.143              NA              NA              NA
## Batch14_2024.144              NA              NA              NA
## Batch14_2024.145              NA              NA              NA
## Batch14_2024.146              NA              NA              NA
## Batch14_2024.147              NA              NA              NA
## Batch14_2024.148              NA              NA              NA
## Batch14_2024.149              NA              NA              NA
## Batch14_2024.150              NA              NA              NA
## Batch14_2024.151              NA              NA              NA
## Batch14_2024.152              NA              NA              NA
## Batch14_2024.153              NA              NA              NA
## Batch14_2024.154              NA              NA              NA
## Batch14_2024.155              NA              NA              NA
## Batch14_2024.156              NA              NA              NA
## Batch14_2024.157              NA              NA              NA
## Batch14_2024.158              NA              NA              NA
## Batch14_2024.159              NA              NA              NA
## Batch14_2024.160              NA              NA              NA
##                  Sample_Gas3_Peak Sample_Gas3_ppm
```

```
## Batch12_2024.1                  NA               NA
## Batch12_2024.2                  NA               NA
## Batch12_2024.3                  NA               NA
## Batch12_2024.4                  NA               NA
## Batch12_2024.5                  NA               NA
## Batch12_2024.6                  NA               NA
## Batch12_2024.7                  NA               NA
## Batch12_2024.8                  NA               NA
## Batch12_2024.9                  NA               NA
## Batch12_2024.10                 NA               NA
## Batch12_2024.11                 NA               NA
## Batch12_2024.12                 NA               NA
## Batch12_2024.13                 NA               NA
## Batch12_2024.14                 NA               NA
## Batch12_2024.15                 NA               NA
## Batch12_2024.16                 NA               NA
## Batch12_2024.17                 NA               NA
## Batch12_2024.18                 NA               NA
## Batch12_2024.19                 NA               NA
## Batch12_2024.20                 NA               NA
## Batch12_2024.21                 NA               NA
## Batch12_2024.22                 NA               NA
## Batch12_2024.23                 NA               NA
## Batch12_2024.24                 NA               NA
## Batch12_2024.25                 NA               NA
## Batch12_2024.26                 NA               NA
## Batch12_2024.27                 NA               NA
## Batch12_2024.28                 NA               NA
## Batch12_2024.29                 NA               NA
## Batch12_2024.30                 NA               NA
## Batch12_2024.31                 NA               NA
## Batch12_2024.32                 NA               NA
## Batch12_2024.33                 NA               NA
## Batch12_2024.34                 NA               NA
## Batch12_2024.35                 NA               NA
## Batch12_2024.36                 NA               NA
## Batch12_2024.37                 NA               NA
## Batch12_2024.38                 NA               NA
## Batch12_2024.39                 NA               NA
## Batch12_2024.40                 NA               NA
## Batch12_2024.41                 NA               NA
## Batch12_2024.42                 NA               NA
## Batch12_2024.43                 NA               NA
## Batch12_2024.44                 NA               NA
## Batch12_2024.45                 NA               NA
## Batch12_2024.46                 NA               NA
## Batch12_2024.47                 NA               NA
## Batch12_2024.48                 NA               NA
## Batch12_2024.49                 NA               NA
## Batch12_2024.50                 NA               NA
## Batch12_2024.51                 NA               NA
## Batch12_2024.52                 NA               NA
## Batch12_2024.53                 NA               NA
## Batch12_2024.54                 NA               NA
```

```
## Batch12_2024.55                 NA              NA
## Batch12_2024.56                 NA              NA
## Batch12_2024.57                 NA              NA
## Batch12_2024.58                 NA              NA
## Batch12_2024.59                 NA              NA
## Batch12_2024.60                 NA              NA
## Batch12_2024.61                 NA              NA
## Batch12_2024.62                 NA              NA
## Batch12_2024.63                 NA              NA
## Batch12_2024.64                 NA              NA
## Batch12_2024.65                 NA              NA
## Batch12_2024.66                 NA              NA
## Batch12_2024.67                 NA              NA
## Batch12_2024.68                 NA              NA
## Batch12_2024.69                 NA              NA
## Batch12_2024.70                 NA              NA
## Batch12_2024.71                 NA              NA
## Batch12_2024.72                 NA              NA
## Batch12_2024.73                 NA              NA
## Batch12_2024.74                 NA              NA
## Batch12_2024.75                 NA              NA
## Batch12_2024.76                 NA              NA
## Batch12_2024.77                 NA              NA
## Batch12_2024.78                 NA              NA
## Batch12_2024.79                 NA              NA
## Batch12_2024.80                 NA              NA
## Batch14_2024.81                 NA              NA
## Batch14_2024.82                 NA              NA
## Batch14_2024.83                 NA              NA
## Batch14_2024.84                 NA              NA
## Batch14_2024.85                 NA              NA
## Batch14_2024.86                 NA              NA
## Batch14_2024.87                 NA              NA
## Batch14_2024.88                 NA              NA
## Batch14_2024.89                 NA              NA
## Batch14_2024.90                 NA              NA
## Batch14_2024.91                 NA              NA
## Batch14_2024.92                 NA              NA
## Batch14_2024.93                 NA              NA
## Batch14_2024.94                 NA              NA
## Batch14_2024.95                 NA              NA
## Batch14_2024.96                 NA              NA
## Batch14_2024.97                 NA              NA
## Batch14_2024.98                 NA              NA
## Batch14_2024.99                 NA              NA
## Batch14_2024.100                NA              NA
## Batch14_2024.101                NA              NA
## Batch14_2024.102                NA              NA
## Batch14_2024.103                NA              NA
## Batch14_2024.104                NA              NA
## Batch14_2024.105                NA              NA
## Batch14_2024.106                NA              NA
## Batch14_2024.107                NA              NA
## Batch14_2024.108                NA              NA
```

```
## Batch14_2024.109                    NA                 NA
## Batch14_2024.110                    NA                 NA
## Batch14_2024.111                    NA                 NA
## Batch14_2024.112                    NA                 NA
## Batch14_2024.113                    NA                 NA
## Batch14_2024.114                    NA                 NA
## Batch14_2024.115                    NA                 NA
## Batch14_2024.116                    NA                 NA
## Batch14_2024.117                    NA                 NA
## Batch14_2024.118                    NA                 NA
## Batch14_2024.119                    NA                 NA
## Batch14_2024.120                    NA                 NA
## Batch14_2024.121                    NA                 NA
## Batch14_2024.122                    NA                 NA
## Batch14_2024.123                    NA                 NA
## Batch14_2024.124                    NA                 NA
## Batch14_2024.125                    NA                 NA
## Batch14_2024.126                    NA                 NA
## Batch14_2024.127                    NA                 NA
## Batch14_2024.128                    NA                 NA
## Batch14_2024.129                    NA                 NA
## Batch14_2024.130                    NA                 NA
## Batch14_2024.131                    NA                 NA
## Batch14_2024.132                    NA                 NA
## Batch14_2024.133                    NA                 NA
## Batch14_2024.134                    NA                 NA
## Batch14_2024.135                    NA                 NA
## Batch14_2024.136                    NA                 NA
## Batch14_2024.137                    NA                 NA
## Batch14_2024.138                    NA                 NA
## Batch14_2024.139                    NA                 NA
## Batch14_2024.140                    NA                 NA
## Batch14_2024.141                    NA                 NA
## Batch14_2024.142                    NA                 NA
## Batch14_2024.143                    NA                 NA
## Batch14_2024.144                    NA                 NA
## Batch14_2024.145                    NA                 NA
## Batch14_2024.146                    NA                 NA
## Batch14_2024.147                    NA                 NA
## Batch14_2024.148                    NA                 NA
## Batch14_2024.149                    NA                 NA
## Batch14_2024.150                    NA                 NA
## Batch14_2024.151                    NA                 NA
## Batch14_2024.152                    NA                 NA
## Batch14_2024.153                    NA                 NA
## Batch14_2024.154                    NA                 NA
## Batch14_2024.155                    NA                 NA
## Batch14_2024.156                    NA                 NA
## Batch14_2024.157                    NA                 NA
## Batch14_2024.158                    NA                 NA
## Batch14_2024.159                    NA                 NA
## Batch14_2024.160                    NA                 NA
```

## PPM to flux

Either using peak2ppm() output data frame or a user created data frame with that same structure, this function converts a time series of gas concentrations in ppm into flux in mg m-2 hr-1. Calculated fluxes correspond to slopes of linear models fit to each set of concentrations sampled over time. Additionally, for cases with 4 sampled concentrations in time ("Timesteps"), the function allows correcting these calculated fluxes fitting four alternative drop-one-step models (each removing a single concentration). The function then returns as "_flux_corrected" the model (original or alternative) with highest $R^2$ (above a 0.7 threshold or that defined by the user in the "Threshold" argument).

```
Final_data <- ppm2flux(sample_ppm, CH4_mass = 16, N2O_mass = 44, Timesteps = 4, Diagnostics = FALSE)
```

```
sample_ppm_check <- read.csv("../data/Input_samples_ppm.csv")

check_data <- ppm2flux(sample_ppm_check, CH4_mass = 16, N2O_mass = 44, Timesteps = 4, Diagnostics = FALS

#generally looks okay!
```

## flux to daily and cumulative emissions

This function uses fluxes calculated with ppm2flux() function as input data frame to calculate: (i) daily fluxes in mg m-2 hr-1 in between sampling events using linear interpolation, (ii) daily emissions in kg ha-1 calculated from those daily fluxes, and (iii) cumulative emissions in kg ha-1 for all target gases. Results for i and ii are summarized within a new "daily_" data frame, as those for iii within an "acc_" data frame.

```
# Load dependencies
library(zoo)
```

```
##
## Attaching package: 'zoo'

## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```
library(tidyr)
library(stringr)
library(lubridate)
```

```
##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union
```

```
flux2acc(check_data)
```

## Cumulative emissions to Global Warming Potential (GWP)

Using cumulative emissions calculated previously with flux2acc(), this function uses the "acc_" data frame to calculate GWP per treatment in $CO_{2-eq}$ for the complete sampling period. $CO_{2-eq}$ for $CH_4$ and $N_2O$ are obtain from IPCC (2021) values for a 100 year time horizon. Nevertheless, they can be modified by users through the CH4_eq and N2O_eq arguments.

```
check_data_GWP <- acc2GWP(acc_check_data)
```